



OPEN Multidimensional social signature de-anonymizes low-sensitivity data

Weiwen Jia^{1,2,5}, Bin Zhou^{3,5}, Xin Lu⁴✉ & Xiaoke Xu^{1,2}✉

Although privacy protection is increasingly important in the digital age, current measures predominantly focus on safeguarding high-sensitivity data, often overlooking the risks associated with low-sensitivity data. Here, we utilize low-sensitivity anonymous interaction data, which depict the behavioral interaction patterns between users and their contacts, to construct multidimensional social signature for identifying users in anonymous datasets. We investigate the potential of low-sensitivity social signature for user identification and propose a classification framework for measuring feature sensitivity levels. Among the test datasets, the accuracy of user identification can reach up to 87% in email communication, and remains wide applicability across different datasets. This suggests that even anonymous low-sensitivity interaction data should be considered personal data warranting protection under existing data protection regulations. We challenge the efficacy of current data anonymization methods and offer new perspectives on low-sensitivity data privacy protection through our feature sensitivity classification framework.

The rapid proliferation of artificial intelligence and big data technologies has dramatically transformed data collection and analysis. While these advancements offer numerous benefits, they also introduce significant risks to privacy. The increased complexity of data processing and the potential for large-scale data breaches have heightened public concerns about the protection of personal information^{1,2}. In response, many countries have enacted laws aimed at protecting personal data pertaining to identified or identifiable individuals. Notable examples include the European Union's General Data Protection Regulation (GDPR)³ and California's Consumer Privacy Act (CCPA)⁴. These regulations establish stringent standards for data processing and protection to address data breaches and privacy issues. According to GDPR and CCPA, anonymous data-processed to make the individual unidentifiable-is not considered personal data, allowing its use, sharing, and sale without the individual's consent. However, even anonymous demographic information^{5,6} or mobile trajectory data⁷⁻¹¹ can potentially enable the re-identification of individuals through dataset linkage, thereby exposing personal information¹². The above data falls within the special categories outlined in current data protection laws, which also encompass biometric data, religious beliefs, medical and health information, and financial account data. These explicitly mentioned categories are widely recognized as high-sensitivity data due to the significant threat their mishandling or disclosure poses to the fundamental rights of individuals. Given the potential risks and legal implications associated with high-sensitivity data, there is heightened concern within the community regarding its protection¹³⁻¹⁵.

In contrast, existing data protection laws often presume that low-sensitivity data poses limited risk to individuals' rights if leaked, leading to insufficient attention to its potential privacy threats. Yet due to its ease of access, such data-like email metadata, Bluetooth proximity logs, or app engagement traces-has been widely collected and shared through academic releases, public APIs, or third-party analytics. Although individually innocuous, these behavioral traces can accumulate over time and expose sensitive patterns related to financial status, interpersonal ties, or lifestyle preferences¹⁶⁻²⁰, creating risks of blackmail, surveillance, and identity theft⁶. Even in anonymized form, the structured and longitudinal nature of such data allows external observers-without platform-level privileges-to re-identify users. Crețu et al.² highlighted this potential through individual-level interaction signals, but overlooked higher-order social patterns between users and their contacts. Building on this insight, we argue that such structural features further amplify privacy risks and warrant renewed attention in both regulatory and technical contexts.

To comprehensively capture interaction patterns between users and their contacts, thereby fully exploiting the identification potential of low-sensitivity data, we propose the concept of multidimensional social signature

¹School of Journalism and Communication, Beijing Normal University, Beijing 100875, China. ²Computational Communication Research Center, Beijing Normal University, Zhuhai, Guangdong 519087, China. ³School of Management Science and Engineering, Nanjing University of Information Science and Technology, Nanjing, Jiangsu 210044, China. ⁴College of Systems Engineering, National University of Defense Technology, Changsha, Hunan 410073, China. ⁵These authors contributed equally: Weiwen Jia and Bin Zhou. ✉email: xin.lu.lab@outlook.com; xuxiaoke@foxmail.com

in this study. This innovative concept describes users' social behavior characteristics from multiple perspectives, thereby enhancing user uniqueness and improving identification accuracy. Specifically, we construct social signatures based on eight characteristics across three dimensions commonly used in social network studies, namely interaction intensity^{21–25}, structural properties^{25–28}, and temporal patterns^{29–31}. We explore the potential of these social signatures in ranking-based^{2,32,33} and threshold-based^{34,35} user identification scenarios. Furthermore, we employ a machine learning model to integrate multidimensional social signature to enhance identification performance. Our results show that the multidimensional social signature achieves up to 87% accuracy in user identification for email communication and maintains high accuracy across different types of datasets. Additionally, to verify the superiority of our approach, we compare it with a notable existing low-sensitivity recognition method currently available. The results show that our approach significantly outperforms this method in terms of accuracy. Finally, we investigate the temporal stability of the multidimensional social signature, finding that its recognition performance remains stable over time, indicating its reliability for long-term use. Overall, we demonstrate that even anonymous low-sensitivity data can be effectively utilized for user identification, emphasizing the urgency of protecting such data. To this end, we develop a framework to categorize feature sensitivity levels, providing new insights into the protection of low-sensitivity data by evaluating the associated risks of these signature features.

Results

The existence of social signatures

Figure 1 presents the distribution of social signature self-distances (d_{self}) and reference distances (d_{ref}) for the Email dataset, showing that individuals maintain highly consistent social signatures over time compared to others within the same period. Statistical tests confirm that d_{self} is significantly smaller than d_{ref} (Supplementary Table 1). This pattern consistently holds across other datasets (Supplementary Figures 1–9, Tables 2–10), thereby validating the presence of distinctive and stable social signatures.

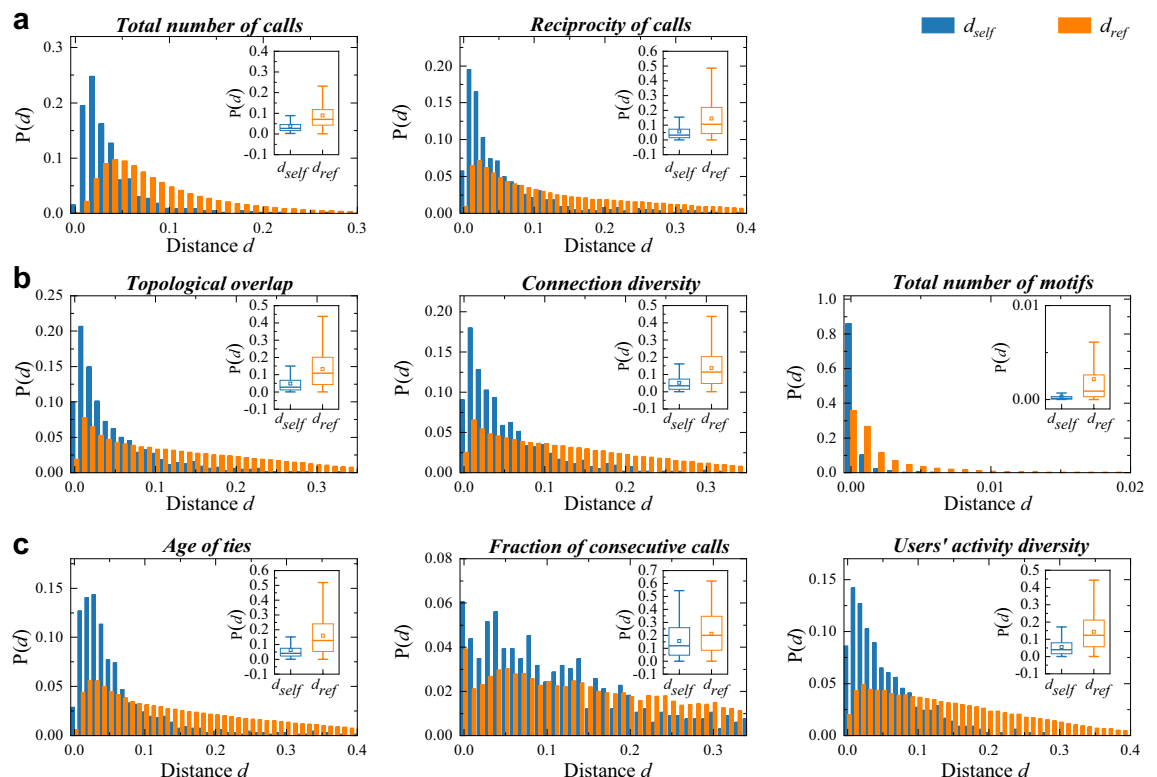


Fig. 1. Distribution of social signature distance. (a) Distribution of self-distance and reference distance specifically for the features of Total number of calls and Reciprocity of calls within the intensity dimension of social signatures. (b) Distribution of self-distance and reference distance for the features of Topological overlap, Connection diversity, and the Total number of motifs within the structural dimension of social signatures. (c) Distribution of self-distance and reference distance for the Age of ties, Fraction of consecutive calls, and Users' activity diversity within the temporal dimension of social signatures. Blue and orange bars show the distributions of self-distance (d_{self}) and reference distance (d_{ref}), respectively. Embedded box plots provide a statistical summary of these distances.

Ranking-based user identification by social signatures

We evaluate the effectiveness of social signature features using ranking-based identification accuracy (Acc_r), which reflects the proportion of users whose correct label is ranked within the top R candidates. Specifically, we measure this accuracy by calculating the proportion of users correctly identified within the top R candidates. As shown in Fig. 2 and Supplementary Figure 10, accuracy increases with higher R , and this trend is consistently observed across datasets, confirming the robustness of our method.

Compared to the bandicoot features proposed by Crețu et al.², our social signature features achieve markedly superior performance. Notably, the “Total number of calls” feature alone surpasses 0.7 accuracy when $R = 100$, significantly outperforming all bandicoot features, which remain below 0.5. Most other social signature features also exceed 0.5 at this rank, demonstrating their strong discriminative power.

Threshold-based user identification by social signatures

We also evaluate user identification effectiveness using threshold-based accuracy (Acc_t), which measures the proportion of correct identifications when comparing unknown users to known users across varying thresholds. Specifically, we determine whether an unknown user shares the same label as a known user by comparing their social signature distance to the threshold θ . As shown in Fig. 3(a) and Supplementary Figures 11–13, accuracy initially rises and then declines as the threshold increases, due to a trade-off between false negatives and false positives. The optimal threshold typically lies near the point of minimal overlap between self-distance and reference-distance distributions—often close to their median—and yields the highest identification accuracy.

Table 1 summarizes the identification accuracy and associated standard deviation for each social signature feature across all datasets at the optimal thresholds. The results demonstrate consistently high identification accuracy under these optimal thresholds, particularly in the email dataset, where most accuracies approach or exceed 0.7. Specifically, the optimal identification accuracies across datasets range from 0.57 to 0.75, with low standard deviations, demonstrating robust threshold-based identification.

As with ranking-based results, our social signature features outperform the bandicoot baseline². Figure 3(b) summarizes the recognition accuracies at the optimal thresholds for social signature features and the bandicoot features on the Bluetooth dataset. The results indicate that under optimal thresholds, most of our features exceed 0.65 accuracy, while the bandicoot features generally fall below that mark, confirming the superior discriminative power of our approach.

User identification based on multidimensional social signature

To further enhance user identification accuracy, we employ the XGBoost algorithm to integrate multidimensional social signature features. Figure 3(b) illustrates that the multidimensional social signature significantly enhances recognition performance on the Bluetooth dataset compared to using single social signature features. This improvement is consistently observed across other datasets (Table 1). Notably, the identification accuracy reaches 0.87 for the Email dataset, while the accuracy for other datasets ranges from 0.74 to 0.85. These findings demonstrate the effectiveness of leveraging multidimensional features for user identification tasks.

We further analyze the contribution of each feature dimension. Structural features yield the highest overall and peak accuracies, followed by temporal features, which show consistent performance across datasets. Intensity features perform well in specific contexts but are generally less robust than the other two dimensions.

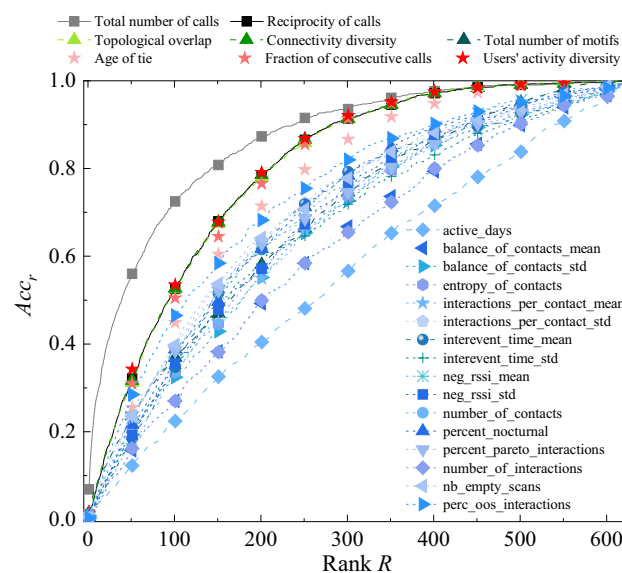


Fig. 2. User identification accuracy across various social signature features (non-blue lines) and bandicoot features (blue lines) in ranking-based method. The horizontal axis (“Rank R ”) denotes the number of candidate users considered, while the vertical axis (“ Acc_r ”) shows the percentage of correctly identified unknown users.

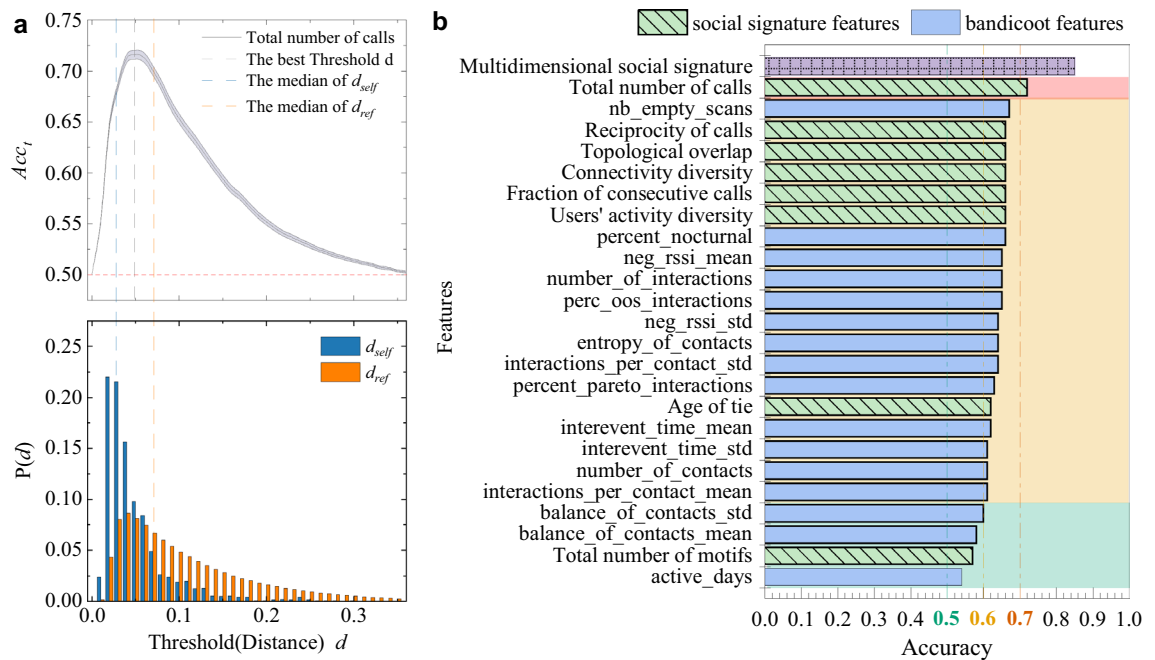


Fig. 3. User identification performance of threshold-based optimization. **(a)** The user identification accuracy of the “Total number of calls” feature across different thresholds. The diagram shows how threshold changes impact accuracy, with blue and orange lines marking the median self-distances and reference distances, respectively. The gray curve plots accuracy against the threshold, highlighting the peak that signifies the optimal threshold for maximizing accuracy. **(b)** Recognition accuracies at optimal thresholds for social signature features and bandicoot features on the Bluetooth dataset. Among them, green bars represent social signature features, blue bars represent bandicoot features, and the purple bar depicts the multidimensional social signature feature. The entire area is divided into three regions—red, orange, and green—indicating high, medium, and low-sensitivity features, respectively.

Type	Feature	Email	Wiki	Facebook	Bluetooth	AskUbuntu			MathOverflow		
						a2q	c2a	c2q	a2q	c2a	c2q
Intensity	<i>N</i>	0.72±0.01	0.62±0.01	0.57±0.01	0.71±0.00	0.63±0.01	0.59±0.01	0.58±0.01	0.57±0.01	0.63±0.01	0.64±0.01
	<i>R</i>	0.67±0.01	0.58±0.01	0.54±0.01	0.66±0.00	0.64±0.01	0.56±0.01	0.58±0.01	0.57±0.01	0.59±0.01	0.64±0.01
	<i>All_i</i>	0.75±0.03	0.74±0.03	0.69±0.02	0.76±0.01	0.73±0.03	0.77±0.03	0.72±0.04	0.68±0.02	0.73±0.02	0.73±0.01
Structural	<i>O</i>	0.69±0.01	0.57±0.01	0.57±0.01	0.66±0.01	0.62±0.01	0.54±0.01	0.57±0.01	0.57±0.01	0.59±0.01	0.63±0.01
	<i>K</i>	0.69±0.01	0.58±0.01	0.55±0.01	0.66±0.00	0.63±0.01	0.55±0.01	0.57±0.01	0.57±0.01	0.59±0.01	0.63±0.01
	<i>M</i>	0.75±0.01	0.63±0.01	0.54±0.01	0.57±0.01	0.58±0.01	0.61±0.01	0.63±0.01	0.54±0.01	0.57±0.01	0.54±0.01
	<i>All_s</i>	0.82±0.02	0.77±0.02	0.72±0.02	0.78±0.02	0.76±0.03	0.75±0.04	0.73±0.04	0.71±0.02	0.73±0.02	0.74±0.02
Temporal	<i>T</i>	0.70±0.01	0.58±0.01	0.54±0.01	0.62±0.00	0.60±0.01	0.63±0.01	0.56±0.01	0.54±0.01	0.58±0.01	0.59±0.01
	<i>MU</i>	0.61±0.01	0.57±0.01	0.54±0.01	0.66±0.01	0.51±0.01	0.56±0.01	0.53±0.01	0.51±0.00	0.59±0.01	0.60±0.01
	<i>A</i>	0.70±0.01	0.57±0.01	0.54±0.01	0.66±0.01	0.61±0.01	0.56±0.01	0.58±0.01	0.54±0.01	0.59±0.01	0.63±0.01
	<i>All_t</i>	0.79±0.02	0.74±0.03	0.70±0.02	0.76±0.01	0.76±0.03	0.80±0.04	0.73±0.04	0.67±0.01	0.74±0.02	0.75±0.01
Overall	<i>All</i>	0.87±0.01	0.82±0.02	0.75±0.02	0.85±0.00	0.78±0.03	0.82±0.04	0.74±0.03	0.74±0.02	0.80±0.02	0.80±0.01

Table 1. User identification accuracy: threshold optimization vs. multidimensional feature fusion. *N*: total number of calls; *R*: reciprocity of calls; *All_i*: the fusion of intensity features; *O*: topological overlap; *K*: connectivity diversity; *M*: total number of motifs; *All_s*: the fusion of structural features; *T*: age of tie; *MU*: fraction of consecutive calls; *A*: users’ activity diversity; *All*: the fusion of all features.

To benchmark against prior work, we combine our features with those used by Crețu et al.² Although initial fusion results appeared suboptimal due to a limited feature subset, full integration of the multidimensional social signature significantly improves identification accuracy (Supplementary Table 11). These results validate the superiority of our framework and underscore the privacy risks posed by seemingly innocuous interaction features.

Feature sensitivity classification framework

The above findings reveal that digital signatures from multidimensional perspectives may embed critical identity information even for low-sensitivity data. To evaluate the privacy risks of different features, we introduce a sensitivity classification framework based on identification accuracy: features with accuracy >0.7 are labeled high-sensitivity, $0.6\text{--}0.7$ as medium-sensitivity, and $0.5\text{--}0.6$ as low-sensitivity. These thresholds are adaptable to different privacy contexts.

Applying this framework to the Bluetooth dataset (Fig. 3(b)), the “Total number of calls” feature is high-sensitivity, “Total number of motifs” is low-sensitivity, and the others fall into the medium range. Bandicoot features mostly lie in the low- and medium-sensitivity ranges, indicating that social signature features are generally more privacy-revealing. Sensitivity rankings also vary across datasets: for example, in the Email dataset, half of the social signature features are classified as high-sensitivity (Table 1). Although not explicitly categorized under privacy laws such as GDPR, such features may still pose significant risks of re-identification.

Importantly, we find that combining low- and medium-sensitivity features can substantially boost identification accuracy. In the Facebook dataset, all individual social signature features are low-sensitivity, yet their fusion achieves 0.75 accuracy—well into the high-sensitivity range. Similarly, in the Wiki dataset, combining a low-sensitivity and a medium-sensitivity feature yields a comparable result. These findings highlight that even non-sensitive features can jointly enable high-risk identification, underscoring the need for stricter regulation and monitoring of their use.

Temporal stability of social signature user identification

We evaluate the temporal stability of user identification using both individual and fused social signature features. Following Crețu et al.², we examine how identification accuracy changes with increasing temporal delay D , using a ranking-based evaluation at Rank R covering 30% of users. Our findings show a general decline in identification accuracy over time (Supplementary Figure 14). In threshold-based user identification methods, we observe a general decrease in accuracy over time (Supplementary Figure 15). However, this decrease is less severe than that observed in ranking-based methods. These results suggest that threshold-based user identification by social signatures is more robust to temporal delays, maintaining relatively stable performance across datasets.

Figure 4 further summarizes temporal stability across datasets when using machine learning to integrate multidimensional social signature features. Despite the inconsistent and greatly varying time spans of the datasets, the accuracy of nearly all datasets changes steadily over time. Notably, the MathOverflow-c2a dataset, which spans the longest time period, still achieves high recognition accuracy even after five periods, spanning more than six years, with the accuracy rate decreasing by only about 10%. Where the length of each period for each dataset is in Supplementary Table 12.

Overall, these findings highlight the ability of machine learning–integrated social signatures to maintain stable identification performance across long-term temporal variations, enhancing the robustness and applicability of user identification in evolving digital environments.

Discussion

With the rapid advancement of digital technology and the Internet, data privacy has become a global priority. Traditionally, efforts have focused on safeguarding high-sensitivity data due to its crucial link to personal privacy, resulting in extensive research, legislation, and multi-layered protection frameworks aimed at preventing unauthorized access and disclosure^{36–42}. However, as technology and social networks evolve, seemingly trivial low-sensitivity data like social interactions, likes, and comments can be aggregated through big data and

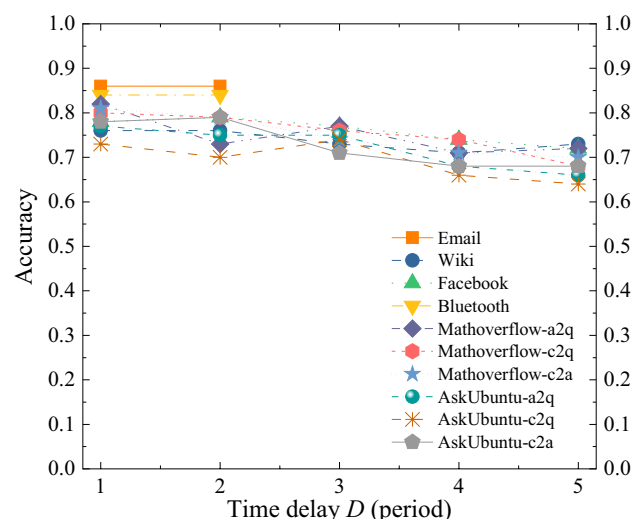


Fig. 4. Temporal stability of multidimensional social signature user identification. It illustrates the variation in user identification accuracy of multidimensional social signature at different time delays across ten datasets.

machine learning to generate detailed personal profiles, revealing users' behavior patterns and interests, and even identifying individuals^{2,43}. Despite this, research on low-sensitivity data remains limited. Our research reveals a sobering reality: low-sensitivity data in anonymous datasets poses a significant identification risk, potentially more threatening than high-sensitivity data due to its accessibility.

In this work, we present the innovative concept of multidimensional social signature to demonstrate the potential of low-sensitivity data for user identification in anonymous datasets. By evaluating 10 real-world datasets, we validate the effectiveness, generality, superiority, and temporal stability of multidimensional social signature in both online and offline interactions. Our findings show that even low-sensitivity anonymous interaction data pose significant privacy risks. However, due to insufficient legal guidance, protection measures for low-sensitivity data are lacking. To address this, we propose a framework that categorizes data sensitivity into high, medium, and low levels, providing clear guidelines for publishing features while ensuring security.

To further evaluate the effectiveness of user identification based on low-sensitivity data, we compare our results with prior studies involving high-sensitivity data. Although ethical and legal constraints prevent a direct comparison using a unified dataset, literature offers valuable benchmarks. For example, just four spatio-temporal points can re-identify 90% of individuals in credit card data and 95% in hourly mobility traces^{7,8}. Similarly, combining ZIP code, date of birth, gender, and number of children can identify 79.4% of Massachusetts residents, and adding attributes like race and occupation increases accuracy to 99.98%⁶. By contrast, our method achieves 87% accuracy using only low-sensitivity data. While this is modestly lower, it underscores the identifiability risks even in the absence of high-sensitivity attributes. Moreover, our model uses a baseline architecture, and performance could improve further with advanced techniques, potentially narrowing the gap.

To better understand these findings, we turn to a key question: why are certain low-sensitivity behavioral features so effective at identifying individuals? Notably, features such as total number of calls and reciprocity consistently rank among the most predictive. We argue that their power stems from deep-rooted cognitive and social constraints. For instance, the total volume of communication reflects inherent limits on human attention and social capacity (e.g., Dunbar's number)^{44,45}. These constraints lead to persistent interaction rhythms and stable tie structures, which form reliable behavioral fingerprints across time²¹. Crucially, the identifiability of these features does not rely on outliers, but rather on consistent within-person regularity—a pattern observed across both online and offline platforms.

Beyond static scenarios, our results also reveal strong temporal robustness. Identification accuracy remains stable even across datasets spanning multiple years, indicating that individual behavioral patterns evolve slowly over time. Combined with prior research on cross-channel social signatures²⁴, this suggests the potential generalizability of our approach to multi-platform or cross-temporal settings. Although our findings suggest temporal stability, validating the generalizability of social signatures across platforms with varying architectures and user behaviors remains an open and essential direction for future research.

However, such generalizability also amplifies privacy risks, as behavioral traces become effective identifiers even beyond their native contexts. To address these emerging risks, we emphasize the need for concrete technical safeguards. While institutional and policy reforms provide the necessary environment, technical mechanisms serve as the first line of defense. Among these, differential privacy (DP) provides mathematically grounded protection by limiting individual exposure during model training. Methods such as DP-SGD^{46,47} introduce calibrated noise to gradient updates, reducing the risk of interaction pattern reconstruction. Meanwhile, data watermarking has emerged as a promising method to trace unauthorized use of user data or model outputs. Recent advances include DWRS⁴⁸, which embeds perturbation-resistant watermarks in sequential recommendation data, and AOW⁴⁹, a black-box watermarking method for protecting model ownership. Text watermarking techniques are also gaining traction in generative models, offering potential applications for large-scale user-generated content⁵⁰. Looking ahead, a hybrid strategy combining DP and watermarking may offer a comprehensive privacy-preserving paradigm that mitigates both information leakage and misuse. We leave the empirical exploration of such combined approaches as promising future work.

In sum, this work broadens the scope of privacy research by demonstrating that low-sensitivity data is far from low-risk. We emphasize the need to re-evaluate data sensitivity hierarchies, prioritize the protection of interactional behavior, and develop both legal and technical tools to safeguard user privacy in increasingly observable digital environments. We believe this work lays a methodological and conceptual foundation for future efforts aimed at safeguarding user privacy in increasingly behaviorally transparent digital systems.

Data and methods

Datasets

Email Dataset. Anonymous internal email communication data from a European research institution, represented as directed edges (u, v, t) where user u sent an email to user v at time t ⁵¹.

Wiki Dataset. Temporal network data capturing Wikipedia users' edits on each other's talk pages, represented as directed edges (u, v, t) where user u edited user v 's talk page at time t ⁵².

Facebook Dataset. Data on Facebook users in the New Orleans area, documenting posts on each other's walls. Directed edges (u, v, t) indicate that user u posted on user v 's wall at time t ⁵³.

Bluetooth Dataset. Close interaction data from over 700 college students collected via Bluetooth over 4 weeks. It includes participant identifiers, timestamps, and received signal strength indicators (RSSI), serving as proxies for device distance^{2,54}.

Ask Ubuntu Series Datasets. These datasets capture user interactions on the Ask Ubuntu subsite of Stack Overflow, divided into three types: AskUbuntu-a2q, with directed edges (u, v, t) indicating user u answered user v 's question at time t ; AskUbuntu-c2q, with directed edges (u, v, t) indicating user u commented on user v 's question at time t ; and AskUbuntu-c2a, with directed edges (u, v, t) indicating user u commented on user v 's answer at time t ⁵¹.

Math Overflow Series Datasets. Similar to the Ask Ubuntu datasets, these datasets record interactions on the Math Overflow subsite of Stack Overflow, categorized as: MathOverflow-a2q, MathOverflow-c2q, and MathOverflow-c2a³¹.

To ensure sufficient user representation in each period, datasets were divided based on their entire time span: long-span datasets into six periods and short-span datasets into three. We applied Li et al.'s activity threshold method²², retaining only users with ten or more contacts per period. Additional details on the datasets and processing methods are in the Supplementary Information.

Experimental setting

We conduct experiments to test the existence of social signatures and their potential for user identification using ranking-based and threshold-based methods. Subsequently, we improve user identification by integrating multidimensional social signature features with the XGBoost algorithm across ten datasets.

In the ranking-based method, we compare the social signatures of unidentified users with those of a known user set to identify a candidate user set whose social signatures most closely resemble those of the unidentified user. For an unidentified user u from the unknown set U'' , we calculate the distance $d(u, v)$ to each known user v from set U' . These distances are ranked, and the top R closest users are selected as candidates. The labels of these candidates form the final set L .

In the threshold-based method, we determine if an unidentified user u from U'' shares the same label as a known user v from U' by comparing their social signature distance against a similarity threshold θ . If $d(u, v) < \theta$, the users are considered identical ($b(u, v) = \text{True}$); otherwise, they are distinct ($b(u, v) = \text{False}$). We use undersampling to balance positive and negative samples and repeat the process across 1000 iterations for robustness.

In the XGBoost algorithm, we balance categories with undersampling and perform 100 randomized trials with an 7:3 training-to-testing data split. We analyze the impact of multidimensional social signature on user identification and propose a framework for classifying feature sensitivity based on feature importance.

We conduct a comparative analysis with Crețu et al., who used low-sensitivity data for user identification via a multilayer perceptron, analyzing behaviors from weekly K -hop network embeddings with features from the 'bandicoot.individual.all' function². We adapt their methodology to incorporate our multidimensional social signature features, focusing on Bluetooth proximity networks. We compare our eight social signature features against sixteen established features from the literature and combine them with those of Crețu et al. for a comprehensive evaluation.

Finally, to assess the social signatures' consistency and accuracy in identifying users over extended periods, despite changing environmental conditions and interpersonal relationships, we evaluate the temporal stability of both individual and fused multidimensional social signature features.

Multidimensional social signature construction

The initial phase of our methodology involves constructing the k -hop individual interaction graph (k -IIG) for any individual in the network, termed the "ego". The k -IIG is a subgraph encompassing the ego and its k -hop neighbors, where " k -hop neighbors" refers to all individuals in the social network who are directly or indirectly connected to the ego through no more than k intermediary links. This graph captures not only the nodes representing these individuals but also all interactions among them. It is crucial to note that while the k -IIG primarily focuses on the interactions between the ego and its direct and indirect contacts, it does not include interactions exclusively among k -hop neighbors. Figure 5 provides illustrative examples of 1-IIG and 2-IIG, demonstrating the structural and relational dynamics within these subgraphs.

Figure 5 outlines the methodology for constructing multidimensional social signatures. Figure 5(a) shows the interaction patterns between the ego and its one- and two-hop neighbors. Figure 5(b) depicts the construction of 1-IIG and 2-IIG, highlighting the interaction intensity and temporal features for direct contacts (1-IIG) and the temporal and structural features for indirect contacts (2-IIG). Figure 5(c) demonstrates the feature extraction process, where mathematical formulas are used to calculate feature values from both 1-IIG and 2-IIG. Finally, Fig. 5(d) details the assembly of an individual's social signature. Specifically, we begin by defining a set of users U containing n individuals, denoted as $U = \{u_1, u_2, \dots, u_n\}$. Concurrently, we specify a collection of time intervals T , comprising m distinct periods, represented as $T = \{t_1, t_2, \dots, t_m\}$. For each user u_i and time interval t_j , we extract a feature vector $x(u_i, t_j)$ that captures the behavioral characteristics of u_i during t_j . Contacts of u_i are then ranked based on the values in $x(u_i, t_j)$, sorted from highest to lowest. The proportional value $p(u_i, t_j)$ of each contact's feature relative to the total feature value is calculated, thereby constructing the social signature $P(u_i, t_j)$ for user u_i in the time interval t_j .

Existence tests for social signatures

To validate the existence of social signatures $P(u_i, t_j)$, we use the Jensen-Shannon Divergence (JSD) to measure the distance between social signatures²¹. JSD effectively quantifies the similarity between two probability distributions, with smaller values indicating greater similarity. The mathematical definition of JSD is as follows:

$$JSD(P_1, P_2) = H\left(\frac{1}{2}P_1 + \frac{1}{2}P_2\right) - \frac{1}{2}[H(P_1) + H(P_2)] \quad (1)$$

where P_1 and P_2 are the two social signatures where $P_i = \{p_i(r)\}$ such that $p_i(r)$ is the fraction of interactions to the contact of rank r in signature i . Additionally, $H(P)$ is the Shannon entropy and $H(P)$ is computed as:

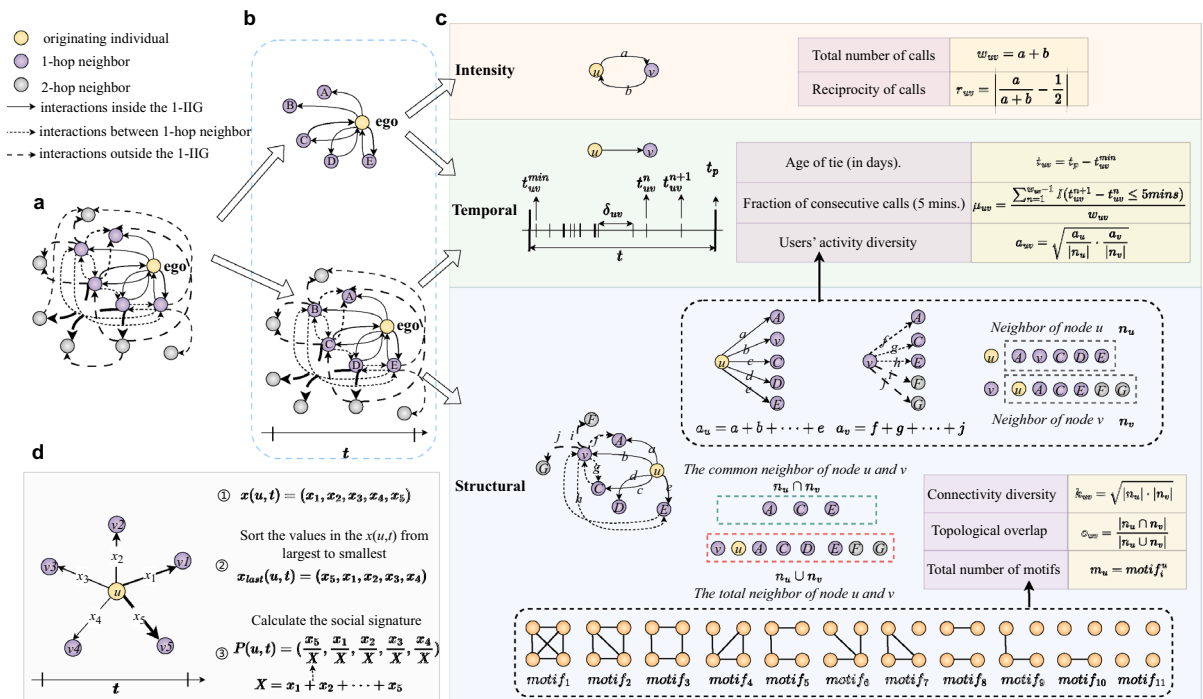


Fig. 5. Multidimensional social signature construction. (a) Displays a localized network for a specific user, featuring the original individual (yellow), 1-hop neighbors (purple), and 2-hop neighbors (gray). Connectivity includes solid lines for direct interactions between the original node and 1-hop neighbors, dotted lines among 1-hop neighbors, and dashed lines connecting 1-hop to 2-hop neighbors and among 2-hop neighbors themselves. All edges are shown as single-directed edges proportional to interaction volume. (b) The top section uses data from the 1-IIG, covering all interactions between the original node and its one-hop neighbors. The bottom section, from the 2-IIG, includes interactions involving the original node, its one-hop, and two-hop neighbors, excluding those only between two-hop neighbors. (c) Demonstrates extraction of multidimensional features from interaction data. (d) Describes the process to create social signatures from the features.

$$H(P) = - \sum_{r=1}^k p(r) \log p(r) \quad (2)$$

where $p(r)$ is as above and k is the maximum value after ranking.

To accomplish this, we initially compute the intra-user distance $d_{\text{self}}(u_i) = JSD(P(u_i, t_j), P(u_i, t_{j+1}))$, which represents the distance between the social signatures of user u_i across consecutive time intervals. Next, we calculate the inter-user reference distance $d_{\text{ref}}(u_i, u_j) = JSD(P(u_i, t_j), P(u_j, t_j))$, which represents the distance between the social signatures of user u_i and other users u_j within the same time interval, where $u_j \in U - \{u_i\}$.

If the distribution of intra-user distances across all periods, $d_{\text{self}} = \{d_{\text{self}}(u_i) : u_i \in U, t_j \in T\}$, is significantly smaller than the inter-user reference distances for the same periods, $d_{\text{ref}} = \{d_{\text{ref}}(u_i, u_j) : u_i, u_j \in U, t_j \in T, u_i \neq u_j\}$, it indicates that users' social signatures are more consistent over time for the same user than between different users. This suggests the existence of distinct social signatures for each user.

Metrics for evaluating the effectiveness of user identification

Ranking-based identification accuracy (Acc_r) is the percentage of users for whom the correct labels are included in the candidate label set, calculated as follows:

$$Acc_r = \frac{\sum_{u \in U''} I(l_u \in L_u)}{|U''|} \quad (3)$$

where l_u is the real label of user u , L_u is the candidate label set for user u , and $I(x)$ is an indicator function that returns 1 if x is true and 0 otherwise.

Threshold-based identification accuracy (Acc_t) is the proportion of correct identifications in comparisons between unknown and known users, calculated as follows:

$$Acc_t = \frac{\sum_{u \in U''} \sum_{v \in U'} I(b(u, v) = (l_u = l_v))}{|U''| \times |U'|} \quad (4)$$

where $b(u, v)$ is a boolean indicating whether users u and v are identified as sharing the same label.

In the XGBoost algorithm, we use Scikit-learn's built-in functions to compute performance metrics. For binary classification, the outcomes are True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). Identification accuracy is defined as the ratio of correctly classified samples to the total number of samples:

$$Acc_s = \frac{TP + TN}{TP + FP + TN + FN} \quad (5)$$

Data availability

Given the data usage agreements between Facebook and Bluetooth, we cannot release the full dataset. Other data used in this study are available in the Stanford Large Network Dataset Collection [<https://snap.stanford.edu/data>].

Code availability

All the code to reproduce the results are available on GitHub (<https://github.com/werjiaweiwen/Social-signature>).

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Author contributions

All authors contributed to the research. W.J. and B.Z. conceived the research. W.J. performed the experiments, and analyzed the data. W.J. and B.Z. wrote the first draft of the manuscript. X.L. and X.X. reviewed and edited the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to X.L. or X.X.

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